

1. Project Overview

In this project, we will provide image labels where we employ Amazon Rekognition. It will be able to recognize and label images. Amazon Rekognition able to identify what it is and label the image.

2. Services used

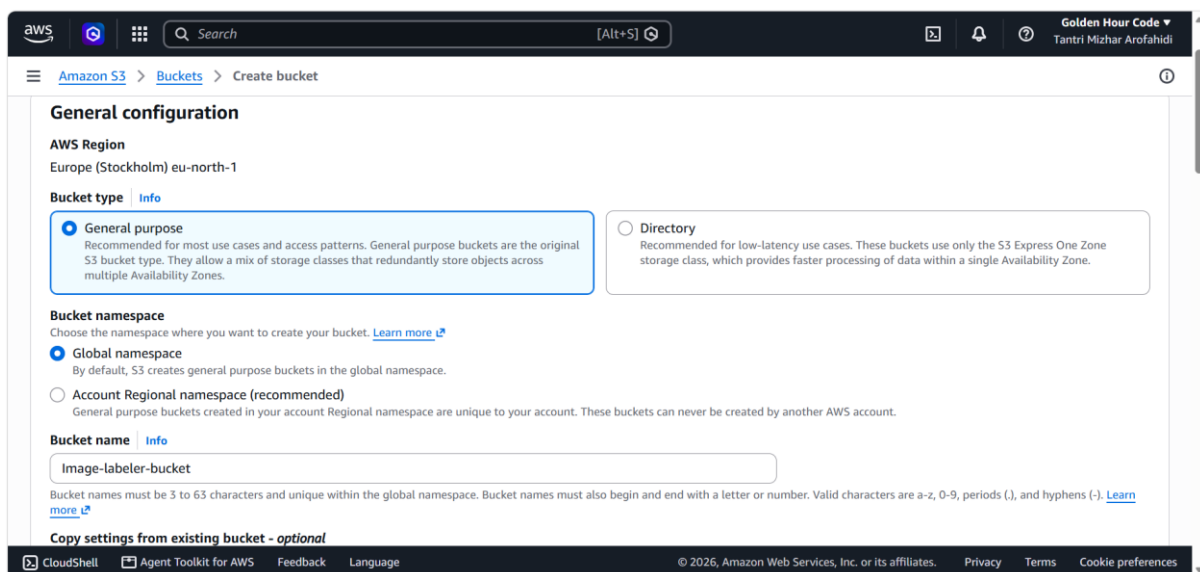
- Amazon S3: Image storage as process of generating labels.
- Amazon Rekognition: Identify and recognize images and generate image labels.
- AWS CLI: Interacting with AWS services through the command-line interface(CLI).

3. Estimated Time & Cost

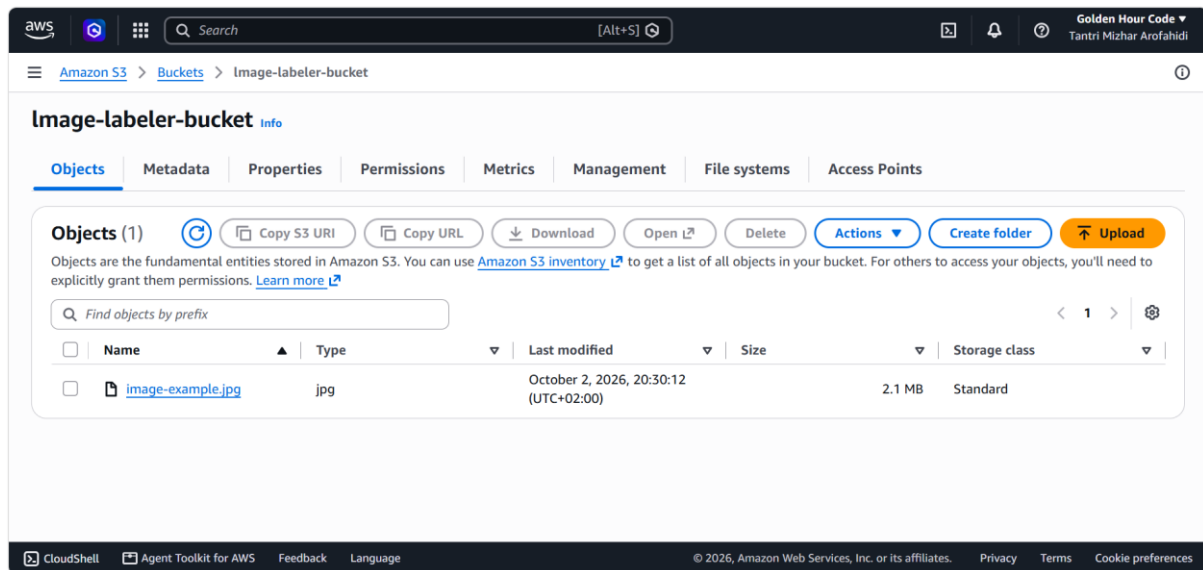
- This project is estimated to take about 20-30 minutes
- Cost: Free (When using the AWS Free Tier)

4. Amazon S3

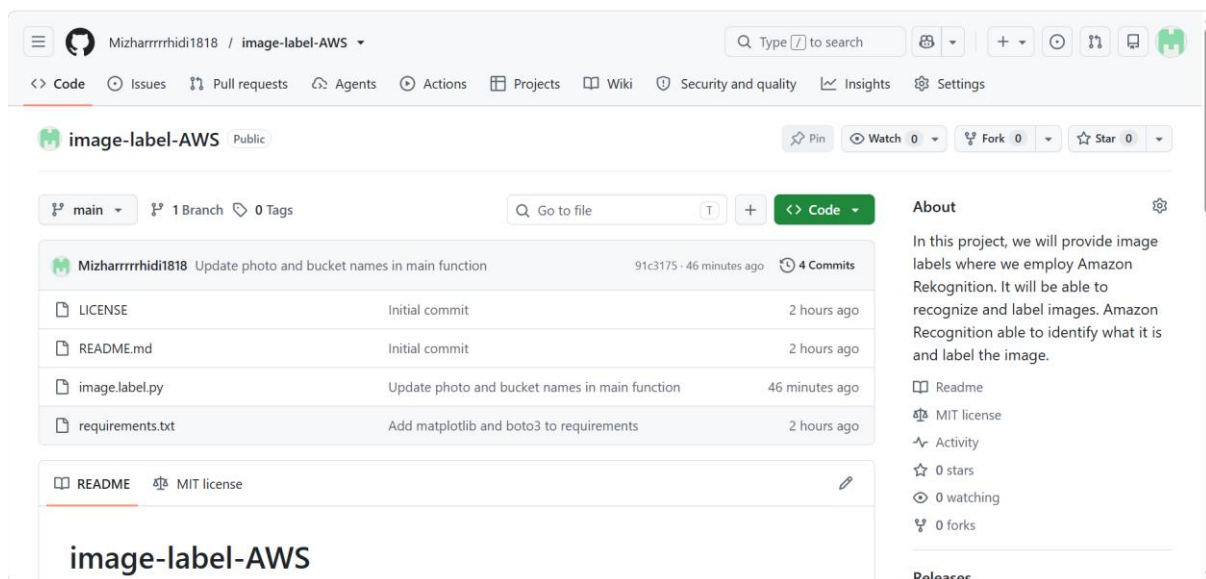
We create a bucket which is its name “image-labeler-bucket”, after that create bucket



After that, we select “image-labeler-bucket” for uploading the image that we want to it.



5. Create a repository on GitHub for connecting the code GitHub with Visual Studio Code



6. Execute AWS CLI (Command Line Interface) in Visual Code

- Let's check on our System that what we have AWS CLI in Command Prompt or PowerShell.

If not, we can install via [Installing or updating to the latest version of the AWS CLI - AWS Command Line Interface](https://awscli.amazonaws.com/v2/install.ps1) Copy + paste "irm https://awscli.amazonaws.com/v2/install.ps1 | iex" on Command Prompt or PowerShell for install

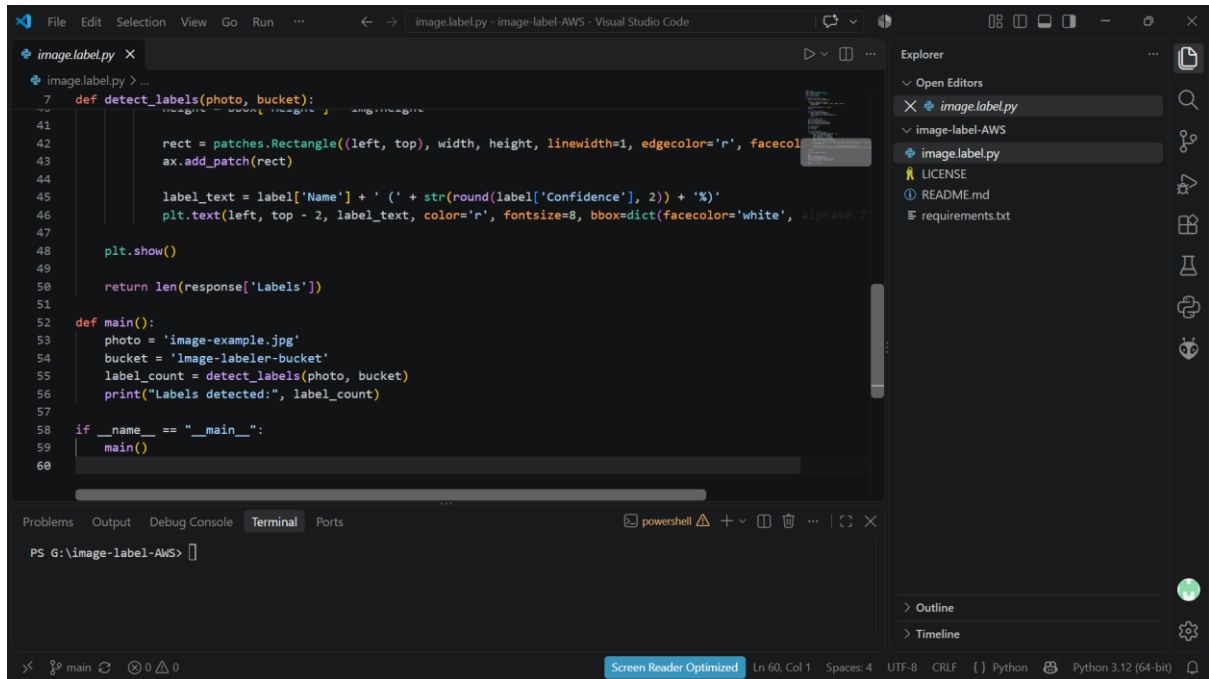
- Check git file for connection with github and visual code studio

<https://github.com/git-for-windows/git/releases/download/v2.56.0.windows.1/Git-2.56.0-64-bit.exe>, then check on command prompt `git --version`

- Connection visual code and Github press `CTRL+SHIFT + P` → Git: Clone <https://github.com/Mizharrrrrhidi1818/image-label-AWS>

This mean to open repository github and execute pyhton code (image.label.py)

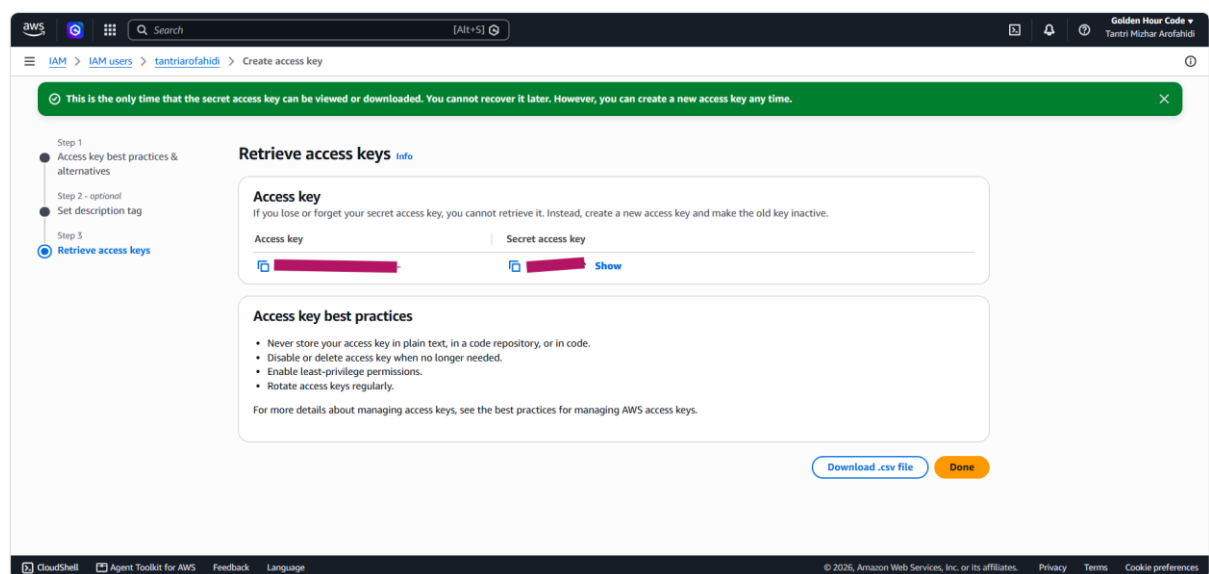
- Open terminal(Command Prompt) with click `CTRL + ``



```
7 def detect_labels(photo, bucket):
41
42     rect = patches.Rectangle(left, top, width, height, linewidth=1, edgecolor='r', facecolor='r')
43     ax.add_patch(rect)
44
45     label_text = label['Name'] + ' (' + str(round(label['Confidence'], 2)) + '%)'
46     plt.text(left, top - 2, label_text, color='r', fontsize=8, bbox=dict(facecolor='white', alpha=0.7))
47
48     plt.show()
49
50     return len(response['Labels'])
51
52 def main():
53     photo = 'image-example.jpg'
54     bucket = 'image-labeler-bucket'
55     label_count = detect_labels(photo, bucket)
56     print("Labels detected:", label_count)
57
58 if __name__ == "__main__":
59     main()
60
```

7. Create an access key on Amazon IAM

We create IAM Users, after that we can know acces key and secret key



Running on visual studio code on terminal write image-label-AWS> aws configure, copy paste access key and secret key

8. Execution python code with python image-label.py

I gain photo labelling from execution python in terminal visual studio code